

# Animal Groups

ABOUT A MILLION different kinds of animals have been discovered and described so far, but there are probably three or four times as many that people have never studied or named. Animals have several features in common.

## Invertebrates

Invertebrates (animals without backbones) were the first animals to evolve on Earth, between 600 and 1,000 million years ago. Hundreds of thousands of species are alive today, and they far outnumber the vertebrates (animals with backbones). Invertebrates come in many different shapes and sizes, including corals, jellyfish, insects, snails, spiders, crabs, centipedes, and worms.

### Characteristic of invertebrates:

- do not have a backbone



Monarch butterfly



Desert tarantula

## Amphibians

Amphibians evolved from fishes more than 350 million years ago. There are more than 7,000 species alive today, including frogs, toads, and salamanders.

### Characteristics of amphibians:

- adults live mainly on land, but breed in water
- cannot maintain a constant body temperature
- skin is usually soft with no scales
- life cycle is usually in three stages: egg, larva (or tadpole), and adult
- tadpoles breathe through gills at first; adults breathe through lungs



Japanese giant salamander



Green toad

## Birds

Birds evolved from reptiles about 140 million years ago. There are more than 10,000 species alive today, including parrots, eagles, penguins, kiwis, owls, and storks. Most birds can fly. They are adapted for flight by having wings instead of front legs, a light skeleton with hollow bones, and a covering of feathers.

### Characteristics of birds:

- birds are the only living animals with feathers
- breathe with lungs
- can maintain a constant body temperature
- lay eggs with hard, waterproof shells; usually incubate eggs with the heat of their bodies



Kiwi



Scarlet macaw

They move, breathe, feed, grow, have young, and respond to changes in their surroundings. To make animals easier to study, biologists divide them into a number of groups. The main groups are shown below.

## Fish

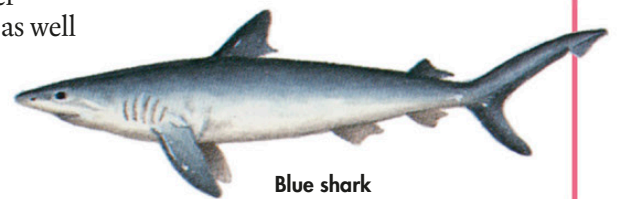
Fish were the first group of vertebrates to evolve from invertebrates about 500 million years ago. There are more than 30,000 species alive today—about the same as all the mammals, birds, reptiles, and amphibians put together. Examples include butterflyfish and sharks.

### Characteristics of fish:

- adapted to live in water
- absorb oxygen from the water through gills; a few have lungs as well
- have fins to help them swim
- bodies are usually covered with scales



Butterflyfish



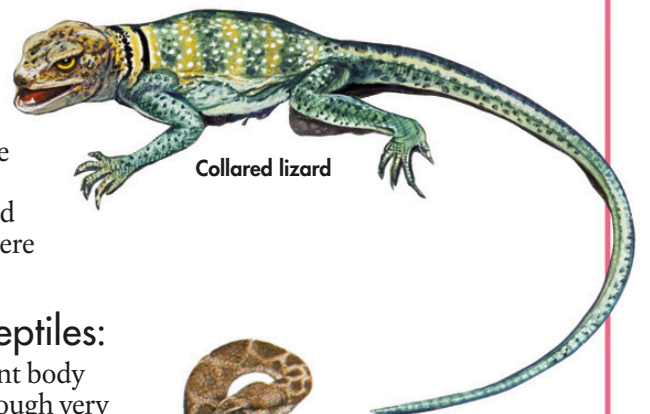
Blue shark

## Reptiles

Reptiles evolved from amphibians about 300 million years ago. More than 10,000 species are alive today, including lizards, snakes, tortoises, turtles, and crocodiles. The dinosaurs were also reptiles.

### Characteristics of reptiles:

- cannot maintain a constant body temperature; may sleep through very hot or very cold weather
- have dry, scaly skin, sometimes with bony plates for protection
- most live and breed on land
- breathe with lungs



Collared lizard



Western diamondback rattlesnake

## Mammals

Mammals evolved from reptiles about 200 million years ago, during the age of the dinosaurs. There are nearly 6,000 species alive today, including kangaroos, rats, cats, elephants, whales, bats, monkeys, and humans.

### Characteristics of mammals:

- mother feeds her young on milk
- bodies are covered with fur or hair
- can maintain a constant body temperature and have sweat glands to cool their bodies
- intelligent, with large brains
- breathe with lungs



Siberian tiger



Kangaroo rat